

Adding Three-Digit Numbers with Regrouping

Answer Key

Grades 2–3

Quick Answers

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| 1. the sum = 475 | 7. (a) the estimate = 900, (b) the exact sum = 913 |
| 2. the sum = 393 | 8. (a) the correct sum = 734, — |
| 3. the sum = 761 | 9. the total books = 500 books |
| 4. (a) the estimate = 800, (b) the exact sum = 801 | 10. (a) the estimate = 900, (b) the exact sum = 843, — |
| 5. the sum = 862 | |
| 6. the total pages = 781 pages | |
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What is verified

8 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 2–3

3.NBT.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 15 tagged checks.

Common wrong answers

3. *If they got 751: regrouped the tens but forgot the carry out of the ones column*
5. *If they got 752: added each column on its own and threw away both carries*
7. *If they got 920: rounded to the nearest ten instead of the nearest hundred*
8. *If they got 624: wrote the ones digit of each column sum and carried nothing*
10. *If they got 856: rounded only the first number and left the second one alone*
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1. Add $347 + 128$.

Solution: Ones: $7 + 8 = 15$, so write 5 and carry one ten. Tens: $4 + 2$ plus the carried ten is 7. Hundreds: $3 + 1 = 4$. One regroup, out of the ones column only. 475

2. Add $256 + 137$.

Solution: Ones: $6 + 7 = 13$, write 3 and carry one ten. Tens: $5 + 3$ plus the carry is 9. Hundreds: $2 + 1 = 3$. 393

3. Add $486 + 275$.

Solution: Ones: $6 + 5 = 11$, write 1 and carry one ten. Tens: $8 + 7$ plus the carry is 16, so write 6 and carry one hundred. Hundreds: $4 + 2$ plus the carry is 7. Two regroupings, one after the other — the second only appears once the first carry has been added in. 761

4. (a) Estimate $412 + 389$ by rounding each to the nearest hundred. (b) Find the exact sum.

Solution (a): 412 is nearer to 400 than to 500; 389 is nearer to 400 than to 300. Adding the rounded numbers gives $400 + 400$. 800

Solution (b): Ones: $2 + 9 = 11$, write 1 and carry. Tens: $1 + 8$ plus the carry is 10, write 0 and carry. Hundreds: $4 + 3$ plus the carry is 8. The exact sum sits one above the estimate, which is what a good estimate should look like. 801

5. Add $594 + 268$.

Solution: Ones: $4 + 8 = 12$, write 2 and carry one ten. Tens: $9 + 6$ plus the carry is 16, write 6 and carry one hundred. Hundreds: $5 + 2$ plus the carry is 8. 862

6. A class read 354 pages in March and 427 pages in April. How many pages in all?

Solution: The question asks for the two amounts combined, so add. Ones: $4 + 7 = 11$, write 1 and carry one ten. Tens: $5 + 2$ plus the carry is 8. Hundreds: $3 + 4 = 7$. 781 pages

7. (a) Estimate $638 + 275$ by rounding each to the nearest hundred. (b) Find the exact sum.

Solution (a): 638 rounds to 600 (the tens digit 3 is below 5) and 275 rounds to 300. Adding the rounded numbers gives $600 + 300$. 900

Solution (b): Ones: $8 + 5 = 13$, write 3 and carry. Tens: $3 + 7$ plus the carry is 11, write 1 and carry. Hundreds: $6 + 2$ plus the carry is 9. The estimate was low by thirteen, so it was a good check. 913

8. Dev added 476 and 258 and wrote 624. (a) Add correctly. (b) Explain what Dev did wrong.

Solution (a): Ones: $6 + 8 = 14$, write 4 and carry one ten. Tens: $7 + 5$ plus the carry is 13, write 3 and carry one hundred. Hundreds: $4 + 2$ plus the carry is 7. 734

Solution (b) — instructor-judged. A correct response names the dropped carries. Full credit for the idea that each column sum over nine had its tens part thrown away instead of carried on: the ten inside the fourteen ones and the hundred inside the twelve tens both went missing, which is why the written answer sits more than a hundred below the true sum. Accept a shorter statement that names one column correctly and shows where its carry should have gone. Saying only that the answer is too small, or that the columns were added wrongly without naming the carry, earns half credit. Restating the correct sum earns no credit, since part (a) already asks for it.

9. A library has 268, 145 and 87 books on three shelves. How many books altogether?

Solution: Three addends, same method, one column at a time. Ones: $8 + 5 + 7 = 20$, write 0 and carry two tens. Tens: $6 + 4 + 8$ is 18, plus the two carried tens is 20; write 0 and carry two hundreds. Hundreds: $2 + 1$ plus the two carried hundreds is 5. A carry can

be more than one when there are three addends.

500 books

10. (a) Estimate $487 + 356$ to the nearest hundred. (b) Find the exact sum. (c) Explain how close the estimate came.

Solution (a): 487 rounds up to 500 and 356 rounds up to 400, so the estimate is $500 + 400$.

900

Solution (b): Ones: $7 + 6 = 13$, write 3 and carry. Tens: $8 + 5$ plus the carry is 14, write 4 and carry. Hundreds: $4 + 3$ plus the carry is 8.

843

Solution (c) — instructor-judged. A correct response compares the two and explains the direction. Full credit for noticing that both numbers were rounded *up*, so the estimate is bound to land above the exact sum — and it does, by fewer than a hundred, which is close enough to catch a badly wrong answer but not close enough to replace the exact one. Accept the difference worked out and used as the reason. Saying only that the estimate is close, with no reason, earns half credit. Claiming the estimate should equal the exact sum earns no credit.